



Exercise Session 8 Highlights

Gathered and Prepared by Farhad Mehdifar

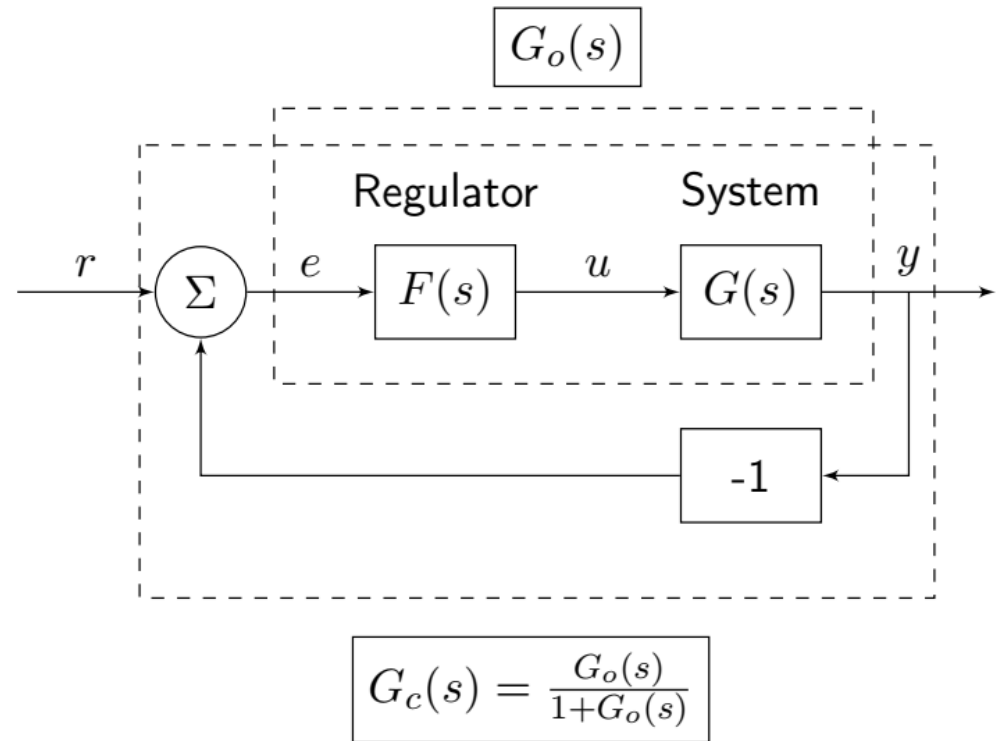
Course: Reglerteknik (EL1000)

2023

Compensator Design

Goals of Compensator Design $F(s)$:

- Achieving a desirable **convergence speed** by means of setting ω_c (gain crossover frequency).
- Achieving desirable **damping** by setting the phase margin φ_m .
- Reducing the **steady-state error** by setting the static gain of the open-loop transfer function, $|G_o(0)|$.



Lead-Lag Compensator:

$$F(s) = K \cdot F_{\text{lead}}(s) \cdot F_{\text{lag}}(s) = K \cdot \underbrace{\frac{\tau_D s + 1}{\beta \tau_D s + 1}}_{\text{lead compensator}} \cdot \underbrace{\frac{\tau_I s + 1}{\tau_I s + \gamma}}_{\text{Lag compensator}}$$

Objective: Tune K , τ_D , β , and γ to achieve desirable closed-loop system performance.

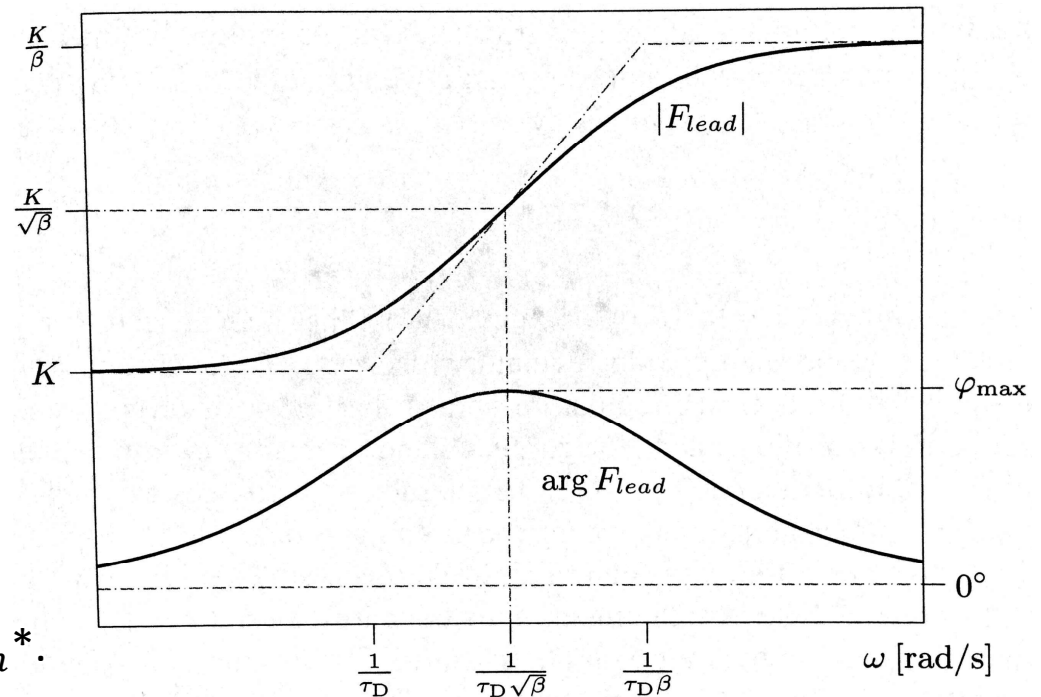
Lead Compensator Design Steps

$$K F_{\text{lead}}(s) = K \frac{\tau_D s + 1}{\beta \tau_D s + 1}$$

Advantage: Improving φ_m and fixing w_c

Design Steps:

1. Determine the desired gain crossover frequency w_c^* .
2. Determine the desired phase margin φ_m^* .



Given φ_m^* determine the necessary *phase shift* (increase of the phase margin) of the open-loop system $G_o(s)$ at w_c^* and set this value to φ_{max} .

Note: if φ_{max} is roughly below 50 degrees one lead compensator is usually enough, otherwise, use cascade-connected lead compensators to avoid imposing a large overshoot (see the “Extra Notes” below for more info).

3. Determine β by $\varphi_{\text{max}} = \arctan \left[\frac{1}{2} \frac{1-\beta}{\sqrt{\beta}} \right]$ alternatively $\left[\arcsin(\eta) = \arctan \left(\frac{\eta}{\sqrt{1-\eta^2}} \right) \right]$
 $\rightarrow \varphi_{\text{max}} = \arcsin \left(\frac{1-\beta}{1+\beta} \right) \Leftrightarrow \beta = \frac{1 - \sin \varphi_{\text{max}}}{1 + \sin \varphi_{\text{max}}}$

Lead Compensator Design Steps (Cont.)

4. Determine τ_D by: $\tau_D = \frac{1}{\sqrt{\beta} \cdot \omega_c^*}$
5. Make sure that the gain crossover frequency of the lead compensated open loop system, i.e., $K F_{lead}(s) G_o(s)$, is equal to ω_c^* by choosing K . In other words:

$$|G(i\omega_c^*)| \cdot \frac{1}{\sqrt{\beta}} \cdot K = 1 \quad \Rightarrow \quad K = \frac{\sqrt{\beta}}{|G(i\omega_c^*)|}$$

Extra Notes on Lead Compensator Design:

- For large a φ_{max} one can design multiple cascade connected lead compensators in which each lead compensator's necessary phase shift effect (φ'_{max}) can be considered as follows:

$$\varphi'_{max} = \frac{\varphi_{max}}{\text{number of cascade connected lead compensators}}$$

- When it is required to design an additional *Lag compensator* (for correcting the steady state error of the closed-loop system) we **add an additional phase of** " $\geq +6^\circ$ " to φ_{max} in step 2. This is because adding a Lag compensator can deteriorate the phase margin of the Lead compensated open loop system.

Lag Compensator Design Steps

$$F_{\text{lag}}(s) = \frac{\tau_I s + 1}{\tau_I s + \gamma}$$

Advantage: increasing the static gain of the open-loop system, thus improving (reducing) the steady-state error of the closed-loop system.

Disadvantage: deteriorates the phase margin!

Design Steps:

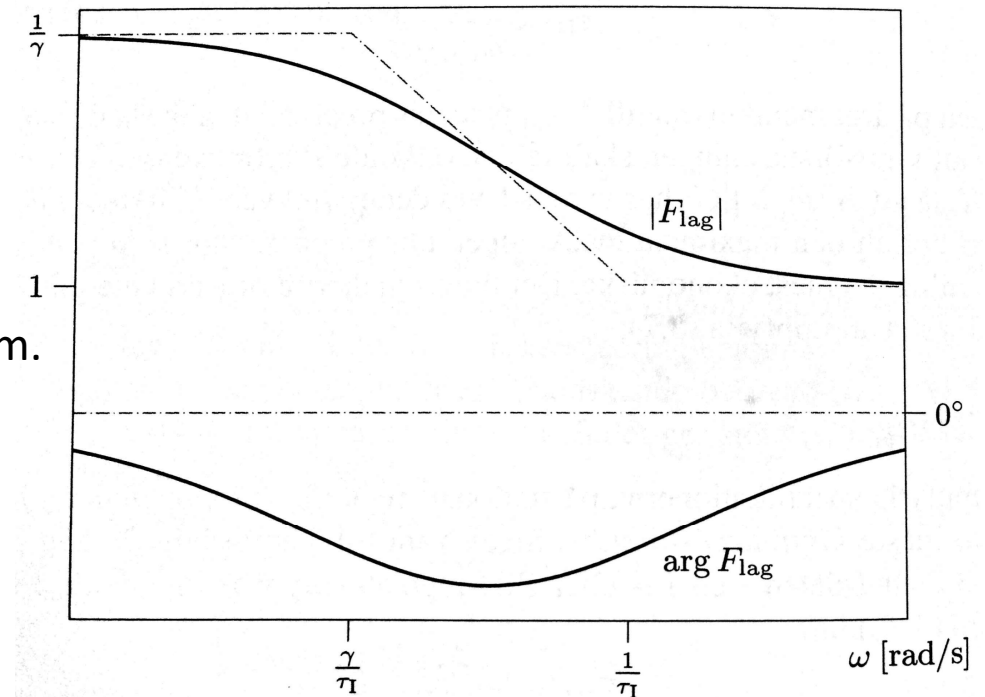
1. Determine the desired steady-state error of the close-loop system e_0 .

2. Compute γ such that:
$$e(t) = \lim_{t \rightarrow \infty} s E(s) R(s) = e_0$$

3. Take $\tau_I = \frac{10}{\omega_c^*}$ (Rule of thumb!)

For this choice we will have: $\arg F_{\text{lag}}(i\omega_c^*) \geq -5.7^\circ$

which justifies the addition of 6 degrees to φ_{max} in step 2 of the lead compensator design.



↑ input

$$E(s) = \frac{1}{1 + \underbrace{F_{\text{lag}}(s) K F_{\text{lead}}(s) G(s)}_{F(s)}}$$

Lag Compensator Design Steps

Extra Notes on Lag Compensator Design:

- If one selects τ_I to be a smaller value then the phase margin of the open loop system will decrease more.
 - if τ_I is too large then the steady-state error will be only reduced for very small frequencies.
-

References

- Lectures of Automatic Control, General Course (EL1000), KTH, 2021
- Adrian Wiltz's notes for Automatic Control, General Course (EL1000), KTH, 2021-2022

Final Note!

*Don't forget to review the notions of **Sensitivity**
and **Robustness Criterion** from classroom
Reglerteknik Lecture 6 Slides!*

Specifically for understanding the sensitivity transfer function better , you can also check **Brian Douglas'** video:



Understanding The Sensitivity Function